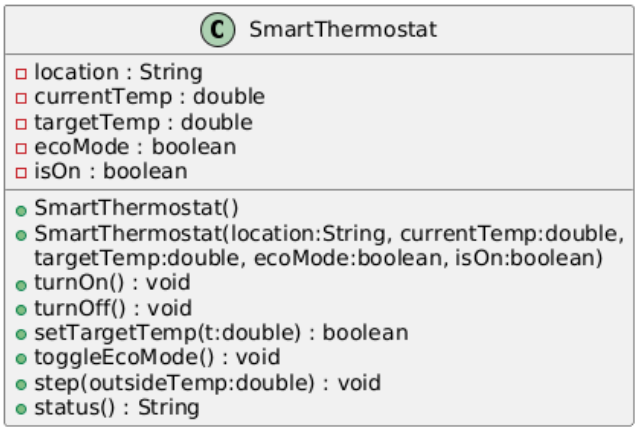
Lab 04: Class and Object

Task 1: SmartThermostat

Goal: model a thermostat with safe updates and simple automation.

Create class `SmartThermostat` with:



- Provide a java code for the above **`SmartThermostat`** Class
- in `main` that runs 20 steps with varying `outsideTemp` and prints status each step.

Task 2: BankAccount + TransactionLog

Goal: objects interacting + bounded history.

Create class `Transaction`:

- Fields: `String type` ("DEPOSIT", "WITHDRAW"), `double amount`, `double balanceAfter`

Create class `TransactionLog`:

- Fields: `Transaction[] entries`, `int size`
- Constructor: fixed capacity (e.g., 50)
- Methods:

`o add(Transaction t)` (ignore if full, or overwrite oldest—choose one and document)

`o Transaction get(int i)` (0..size-1)

`o int count()`

Create class `BankAccount`:

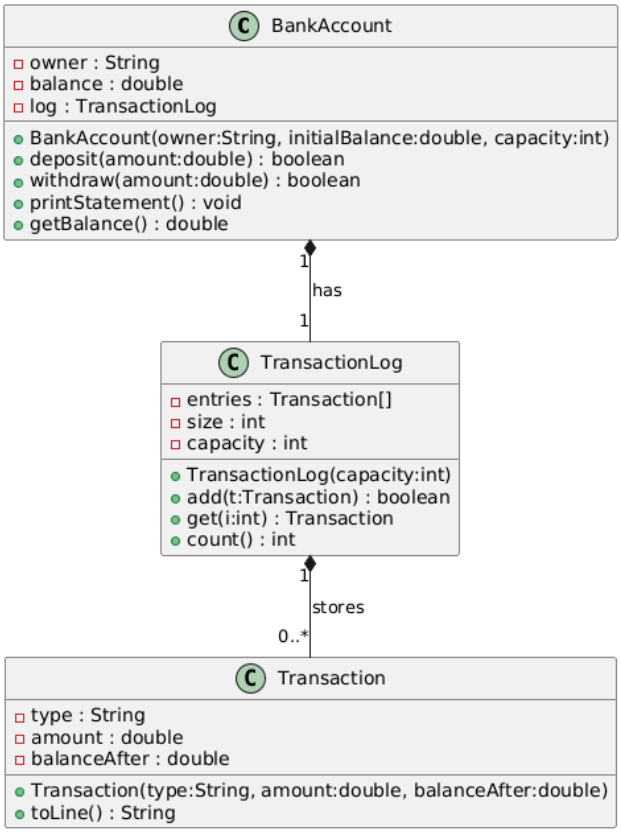
- Fields: `String owner`, `double balance`, `TransactionLog log`
- Constructor: owner, initial balance, log capacity
- Methods:

`o deposit(double amount)` (amount > 0)

`o withdraw(double amount)` (amount > 0 and <= balance)

`o printStatement()` prints all transactions + final balance

Main Class: create account, do 10 operations, print statement.



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Task 3 — LibraryRoom Reservation

Goal: implement logic using plain objects and arrays.

Class `TimeSlot`:

- Fields: `int day` (1..7), `int startHour` (0..23), `int endHour` (1..24)
- Validation: `startHour < endHour`
- Method: `boolean overlaps(TimeSlot other)`

Class `Reservation`:

- Fields: `String studentName`, `TimeSlot slot`
- Method: `String summary()`

Class `StudyRoom`:

- Fields: `String roomName`, `Reservation[] reservations`, `int size`
- Methods:
  - `boolean reserve(String student, TimeSlot slot)`
    Reject if overlaps with existing reservations
  - `boolean cancel(String student, TimeSlot slot)` (match by student + same slot values)
  - `void printSchedule(int day)`

Deliverable:

- Provide full class diagram of the LibraryRoom Reservation
- Provide Java code of that diagram
  - in `main`, add reservations, test overlap rejection, cancel one, re-add.

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